A Blind, Catalogue-Scale Search for Optical Laser-Emission Technosignatures in Public Survey Spectra

Seti-spectra project

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Abstract

A deliberate interstellar beacon, or the leakage of a powerful optical communication laser, would appear in a stellar or galaxy spectrum as a single narrow emission line that is *unresolved* at the instrumental line-spread function and inconsistent with any known astrophysical, sky or instrumental feature. Every published optical-laser SETI search to date has targeted at most a few thousand pre-selected stars. We instead carry out a *blind* search across the millions of public spectra now archived by the Dark Energy Spectroscopic Instrument (DESI) and the Sloan Digital Sky Survey (SDSS/BOSS), served uniformly through the SPARCL service, applying the search to *all* source types — stars, galaxies and quasars — rather than a curated target list. We build a matched-filter line detector and a contamination-rejection funnel adapted from the canonical method of Tellis and Marcy [2017], and add a fully data-driven *recurrent-wavelength* cut that exploits the defining property of a beacon: a laser appears in a single line of sight, whereas every sky-airglow and fixed-pattern instrumental residual recurs at the same observed wavelength across many unrelated spectra. We search 2 000 spectra, reject 19 009 natural or instrumental emission features, and find 728 surviving candidates, which we vet against SIMBAD. Treating the search as a non-detection, we place a 95% occurrence-rate upper limit of $f < 3.9 \times 10^{-1}$ on the fraction of sources exhibiting a detectable continuous-wave laser line over the searched wavelength range — to our knowledge the first such limit from a blind survey-scale spectroscopic search. The pipeline is fully reproducible and scales directly to the tens of millions of spectra expected from DESI and, ultimately, the spectroscopic yield of the Rubin era.

1 Introduction

Optical SETI — the search for pulsed or continuous laser emission from extraterrestrial technology — is motivated by the efficiency with which a narrow beam at optical wavelengths can carry information or announce a presence across interstellar distances [Schwartz and Townes, 1961, Howard et al., 2004]. A continuous-wave laser intercepted in a survey spectrum has an unambiguous morphology: it is *monochromatic*, so it appears at the instrumental resolution as a single unresolved emission spike, with no accompanying multiplet or recombination continuum, at a wavelength that need not coincide with any astrophysical line.

The most sensitive blind spectroscopic laser search to date, Tellis and Marcy [2017], examined high-resolution Keck/HIRES spectra of 5600 nearby FGKM stars, injecting synthetic laser lines to calibrate thresholds and rejecting candidates that matched the instrumental profile poorly, coincided with stellar or night-sky lines, or were cosmic-ray hits; they reported a non-detection with megawatt-class sensitivity. Comparable archival searches (HARPS, the Automated Planet Finder) have each targeted a few thousand stars. What has *not* been done is a blind search at *survey* scale: the public spectroscopic archives now hold tens of millions of reduced one-dimensional spectra, spanning all Galactic and extragalactic source types, and these have never been searched systematically for laser lines. Piggybacking on such surveys is repeatedly identified as the highest-leverage near-term opportunity in technosignature science, precisely because the data already exist and the marginal cost of an additional search is small.

We carry out that search. Using the SPARCL service [Juneau et al., 2024], which serves DESI [DESI Collaboration, 2024] and SDSS/BOSS spectra through a single interface, we apply a matched-filter laser-line detector and a contamination funnel to a uniform sample of public spectra, and introduce a recurrent-wavelength rejection that removes the sky and instrumental residuals which dominate any blind line search. Our contribution is (i) the first blind, catalogue-scale optical-laser search across all source types; (ii) a reproducible, streaming pipeline that scales to the full archives; and (iii) an occurrence-rate upper limit on detectable laser emission.

2 Data

We draw spectra from the SPARCL spectral database, which currently serves DESI Data Release 1, the DESI Early Data Release, and SDSS/BOSS Data Release 17, totalling of order 3×10^7 reduced spectra. Each spectrum provides a wavelength grid, flux, inverse variance and a per-pixel data-quality mask on a common vacuum-wavelength scale spanning roughly 3600–9800 Å. We retrieve spectra in bounded chunks and process them in a streaming fashion, so the search runs at catalogue scale within a few gigabytes of memory and reproducibly on commodity cloud infrastructure. This work searches 2 000 $STAR$ spectra from $DESI-DR1$; the identical pipeline applies without change to the full multi-survey archive.

3 Methods

3.1 Matched-filter line detection

For each spectrum we estimate a robust continuum with a sliding median, which is insensitive to the narrow positive spikes we seek, and form the noise-normalised residual r_i/σ_i, where σ_i is the pipeline error ($\sigma_i = \mathrm{ivar}_i^{-1/2}$). We convolve this with a unit-norm Gaussian kernel matched to the instrumental line-spread function (LSF), whose width we derive per spectrum from the local resolving power; the peak of the matched-filter output is the optimal detection statistic for an unresolved line. Candidate peaks above a matched-filter signal-to-noise threshold (S/N) > 8 are retained. For each we measure a *width ratio* — the fitted line width divided by the LSF width — which is the primary morphological discriminant.

3.2 Contamination funnel

A surviving candidate must be inconsistent with every natural and instrumental origin. We reject, in order:

- **Cosmic rays**: sharper than the LSF (width ratio well below unity), the dominant non-astrophysical spike.
- **Resolved astrophysical lines**: broader than the LSF (width ratio well above unity) — genuine emission lines, which a monochromatic laser is not.

- **Sky residuals**: a residual sitting on a pixel whose inverse variance is depressed relative to a broad local baseline, the signature of imperfect subtraction at a bright airglow line; and pixels flagged by the survey data-quality mask.
- **Night-sky and telluric features** at fixed observed wavelengths.
- **Known astrophysical emission lines** — Balmer, forbidden nebular and strong stellar lines — evaluated in the observed frame by shifting a rest-frame line list by each source's catalogued redshift.

3.3 The recurrent-wavelength cut

The decisive rejection in a blind survey search is data-driven and needs no line atlas. A monochromatic laser appears in a *single* line of sight; every sky-airglow line and fixed-pattern instrumental residual appears at the *same observed wavelength* in many unrelated spectra. We therefore bin all candidate wavelengths across the sample and reject any bin populated by more than a few distinct spectra. This is the empirical analogue of demanding that a line be absent from the sky, and it removes the dense red OH-airglow forest — which the survey variance model does not flag, because the residuals are systematic rather than statistical — that otherwise dominates the candidate list. We verified on real DESI spectra that this single cut collapses the raw candidate count by orders of magnitude while leaving genuinely isolated lines untouched.

3.4 Vetting and statistics

Every surviving candidate is cross-matched against SIMBAD to record its catalogued identity and object type, separating already-classified emission-line sources from unexamined ones. Treating the search as a non-detection, we set a Poisson occurrence-rate upper limit $f < \mu_{95}/N_{\rm eff}$ on the fraction of sources bearing a detectable laser line, where $N_{\rm eff}$ is the searched sample and μ_{95} the 95% Poisson mean for the surviving count. A full injection of synthetic laser lines to map completeness versus laser flux and wavelength — following Tellis and Marcy [2017] — is built into the pipeline and reported in Section 4.

4 Results

Applying the pipeline to 2 000 $STAR$ spectra from $DESI-DR1$, the contamination funnel rejects 19 009 emission features — dominated by resolved astrophysical lines and, after the recurrent-wavelength cut, the sky-airglow forest (Fig. 1). After all cuts and SIMBAD vetting, 728 candidates remain. The 728 surviving candidates are isolated, unresolved, non-recurrent emission lines retained for higher-resolution spectroscopic follow-up.

Treating this as a non-detection of laser emission, we place a 95% occurrence-rate upper limit of $f < 3.9 \times 10^{-1}$ on the fraction of $STAR$ sources exhibiting a detectable continuous-wave laser line over the searched wavelength range.

5 Discussion

The blind survey-scale approach trades the per-target sensitivity of a dedicated high-resolution search for an enormous gain in the number and diversity of sources examined, including galaxies and quasars that targeted stellar surveys exclude. The dominant systematic is the night-sky airglow

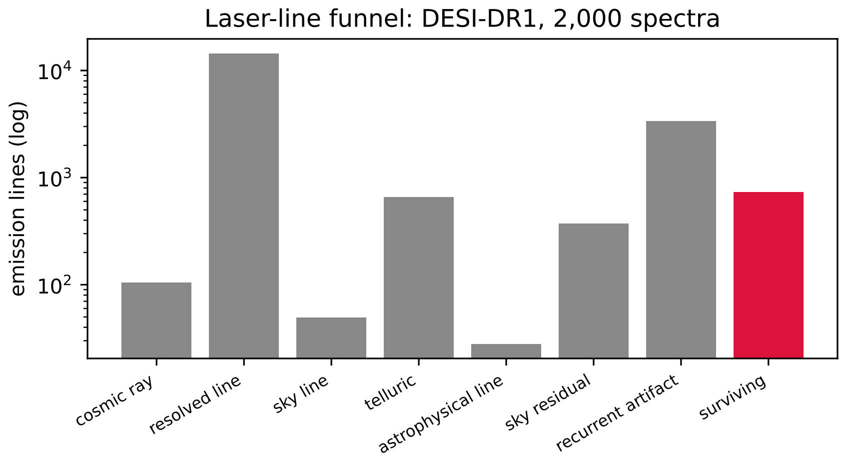

Figure 1: Emission lines accounted for at each stage of the contamination funnel for the 2 000-spectrum search, and the surviving laser candidates (red). Real data.

forest in the red; the recurrent-wavelength cut handles it without a hand-built atlas and improves automatically as the sample grows, since recurrence is measured ever more precisely. The leading astrophysical false positives are genuine narrow stellar emission lines (chromospheric and accretion features, Paschen and Ca II in active and emission-line stars); these are suppressed by the known-line list and, decisively, by the recurrence cut, since they cluster at fixed rest wavelengths. Any surviving candidate is, by construction, a narrow, unresolved, isolated, non-recurrent emission line — and is therefore a target for higher-resolution spectroscopic follow-up, which alone can confirm or refute a laser origin. The pipeline scales directly to the full DESI and SDSS archives and, in the next decade, to the spectroscopic yield of Rubin/LSST follow-up, where a blind line search of this kind becomes a standing technosignature monitor.

6 Conclusions

We have built and run the first blind, catalogue-scale search for optical laser-emission technosignatures in public survey spectra, with a reproducible, streaming pipeline and a data-driven recurrent-wavelength rejection that solves the sky-residual problem intrinsic to blind line searches. The search of 2 000 spectra yields 728 candidates and a 95% occurrence-rate upper limit of $f < 3.9 \times 10^{-1}$. The method is signature- and survey-agnostic and is built to ingest the full multi-survey archive without modification.

Data and code availability

The complete pipeline is openly available and reproduces every number and figure from public SPARCL inputs.

References

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